What Have I Learned About Energy?

Unit 1 Study Guide · Energy Transfer & Transformation · Student

Name: _____ **Date:** _____

Use this sheet to get ready for the Unit 1 assessment.

A · MATCH THE WORD

Write the letter of the correct meaning on the line.

1. energy _____

2. evidence _____

3. stored energy _____

4. energy of motion _____

5. transfer _____

6. transformation _____

7. collision _____

8. criteria _____

9. constraints _____

10. refine _____

Meanings:

A. energy changing form

B. the energy a moving object has

C. the limits (materials, time, safety)

D. energy ready to cause change later

E. what a good solution must do

F. the ability to cause change

G. to improve using test evidence

H. two or more objects coming together

I. energy moving from place to place or object to object

J. what we observe that shows a change occurred

B · FILL IN THE BLANK

Use a word from the bank. Some words may be used more than once.

Word bank: cause change evidence stored motion faster transfer transformation

1. Energy is the ability to _____ .
2. A change we observe is our _____ that energy was involved.
3. A battery, a stretched rubber band, and a raised object all have _____ energy.
4. A moving object has energy of _____ .
5. For the SAME object, moving at a _____ speed means more energy of motion.
6. During a collision, energy of motion can _____ to the target.

C · TRANSFER, TRANSFORMATION, OR BOTH?

Circle the right answer for each one.

1. Warm water warms a cool spoon. The heat moves, and it stays heat.

Circle: **TRANSFER** **TRANSFORMATION** **BOTH**

2. A flashlight battery turns stored energy into light.

Circle: **TRANSFER** **TRANSFORMATION** **BOTH**

3. A domino falls and clicks: motion passes to the next domino AND some becomes sound.

Circle: **TRANSFER** **TRANSFORMATION** **BOTH**

4. A fan changes electric current into spinning motion.

Circle: **TRANSFER** **TRANSFORMATION** **BOTH**

D · SAME OBJECT, DIFFERENT SPEED

For the SAME object: faster speed → more energy of motion → a bigger change.

As ramp height goes up, the cup moves _____ (farther / not as far).

This is evidence that a faster car has _____ energy of motion (more / less).

In the Ramp It Up investigation:

We MEASURE _____ .

We INFER _____ .

E · LABEL THE ENERGY CHANGE

Label the INPUT and the OUTPUT for a flashlight.

INPUT: _____ → **OUTPUT: _____**

F • WHAT HAPPENS IN A COLLISION?

BEFORE: a moving object → **COLLISION** → **AFTER: target moves + sound may happen**

After a collision, energy does NOT _____ .

Instead, the energy of motion _____ to the target, and some becomes sound.

Hint: it does NOT disappear.

G • ENGINEERING WORDS

Criteria = _____ .

Constraints = _____ .

Refine = _____ .

H • EDISON & LATIMER

Edison did NOT invent the first light bulb; he _____ earlier designs.

His team tested more than _____ materials.

In 1879, Edison found a _____ filament with a carbon coating.

Latimer improved the bulb with a stronger, more durable _____ .

I • SHORT RESPONSE

Pick ONE device: a flashlight, a buzzer, or a ramp toy. Write a claim, give evidence, and explain your reasoning. Use "supports," not "proves."

Claim: _____

_____

Evidence: _____

_____

Reasoning: _____

_____